

Nonlinear-Embedding Transformer based Modulation Recognition for Sub-Nyquist Sampling Systems with Reduced Channels

Heping Lu

Hangzhou Institute of Technology
Xidian University
Hangzhou, China
24241215151@stu.xidian.edu.cn

Dawei Gao

Hangzhou Institute of Technology
Xidian University
Hangzhou, China
gaodawei@xidian.edu.cn

Yaxing Yue

Hangzhou Institute of Technology
Xidian University
Hangzhou, China
yueyaxing@xidian.edu.cn

Yufeng Chen

Hangzhou Institute of Technology
Xidian University
Hangzhou, China
chenyufeng@xidian.edu.cn

Liyang Chen

Hangzhou Institute of Technology
Xidian University
Hangzhou, China
chenliyang@xidian.edu.cn

Guisheng Liao

Hangzhou Institute of Technology
Xidian University
Hangzhou, China
liaogs@xidian.edu.cn

Abstract—To address high sampling rates and complex environments in next-generation communications, this paper proposes a novel framework based on block-based nonlinear embedding and polynomial normalization activation function (BNEPN) for joint wideband spectrum sensing and modulation recognition. Leveraging a multi-coset sampling architecture without signal reconstruction, the method ensures robust performance at reduced rates. Specifically, a block-based nonlinear embedding module is proposed to compress signal sequences for enhanced efficiency, while the polynomial normalization activation function is introduced to resolve aliasing-induced nonlinear couplings and prevent feature over-smoothing. Furthermore, composite sub-band identifiers, comprising static encodings and learnable biases, are employed to explicitly integrate spectrum occupancy information. Experimental results demonstrate our proposed BNEPN's effectiveness even below twice the Landau rate, achieving 99.99% sensing accuracy and superior modulation recognition compared to other models.

Index Terms—Deep learning, Spectrum sensing, Modulation recognition, Multi-coset sampling, Landau rate.

I. INTRODUCTION

The transition to 6G exploits higher frequency bands like the terahertz spectrum to expand bandwidth, yet this challenges the Nyquist theorem with prohibitive sampling rate demands [1]. Concurrently, the increasing diversity of modulation types exacerbates the complexity of the electromagnetic environment [2]. Consequently, spectrum sensing and modulation recognition have become indispensable core technologies for accurate target information acquisition.

Conventional narrowband spectrum sensing techniques, such as energy [3], matched filter [4], and cyclostationary feature detection [5], are ill-suited for wideband environments. For multiband sensing, narrowband scanning suffers from low efficiency [6], while wideband approaches [7] depend on costly high-speed analog-to-digital converters (ADCs) [8].

Although compressed sensing can achieve sub-Nyquist sampling [9], practical signals often fail to meet the requisite sparsity conditions [10], [11]. Similarly, traditional modulation recognition methods face limitations: likelihood-based approaches [12] demand extensive prior knowledge [13], whereas feature-based methods [14] rely on laborious manual feature engineering [15]. To mitigate excessive sampling rates, multi-coset sampling (MCS) [16] was introduced. Following the initial application of convolutional neural networks (CNNs) to modulation recognition [17], Wu et al. [18] achieved recognition directly from MCS signals but neglected spectral occupancy. While SigChord [19] incorporates spectral information, its reliance on signal recovery significantly increases computational complexity.

This paper proposes the block-based nonlinear embedding and polynomial normalization activation function (BNEPN) framework for joint wideband spectrum sensing and modulation recognition. The contributions are summarized as follows:

- A reduction in total sampling rate is achieved by minimizing utilized cosets while maintaining robust detection performance.
- A block-based nonlinear embedding (BNE) module is proposed to compress signal sequences, significantly enhancing computational efficiency and training speed.
- The polynomial normalization activation function (Poly-Norm) is introduced to resolve aliasing-induced nonlinear couplings and prevent feature over-smoothing in deep networks.
- A composite subband identifier mechanism, combining static encodings with learnable biases, is designed to explicitly integrate spectrum occupancy information into modulation recognition.

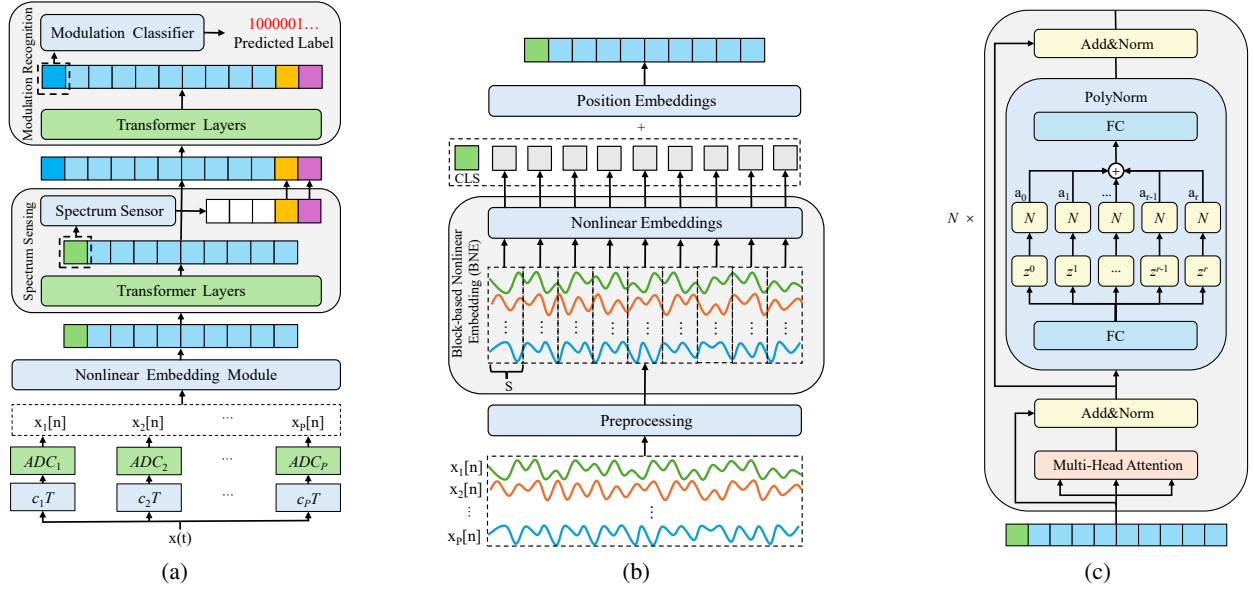


Fig. 1: The overall network architecture of the proposed BNEPN framework for spectrum sensing and modulation recognition. (a) The overall structure of the proposed network. (b) Details of nonlinear embedding module. (c) Details of Transformer layers.

II. SIGNAL MODEL

As shown in Fig. 1(a), the received multi-band signal $x(t)$ can be represented as

$$x(t) = \sum_{i=1}^M h_i(t) * a_i(t) e^{j2\pi f_i t} + n(t), \quad (1)$$

where M is the maximum number of occupied subbands, $h_i(t)$ is the channel response, $*$ denotes convolution, $a_i(t)$ is the transmitted signal in the i -th occupied sub-band, f_i is the carrier frequency, and $n(t)$ is noise.

In this paper, we employ the MCS framework for sub-Nyquist sampling of multi-band signals, which consists of P parallel channels. Each channel includes a delay unit and a low-speed ADC operating at a sampling rate of $F_s = F/L$, where F is the Nyquist sampling rate. The total sampling rate is $F_t = PF_s$. Furthermore, Landau [20] demonstrated that when spectral occupancy is known, the minimum F_t can be reduced to the sum of the occupied bandwidths, which is defined as the Landau rate, F_l . Specifically, we define a set of integers $\{c_q\} = \{c_1, c_2, \dots, c_P\}$, where $0 \leq c_1 < c_2 < \dots < c_P \leq L - 1$. The delay for the q -th branch is $c_q T$, with $T = 1/F$ being the Nyquist sampling period. The sampling sequence of each branch can be expressed as

$$x_q[n] = x(nT), n = mL + c_q, m = 0, 1, 2, \dots, \quad (2)$$

which constitutes the input to the network model, denoted as

$$\mathbf{x}[n] = [x_1[n], x_2[n], \dots, x_P[n]]^T. \quad (3)$$

Then, accumulated N discrete samplings, the complex-valued signal $\mathbf{X} = [\mathbf{x}[1], \dots, \mathbf{x}[N]] \in \mathbb{C}^{P \times N}$ is expanded and transposed into a real-valued signal $\mathbf{X}' \in \mathbb{R}^{N \times 2P}$.

Our objective is to further downsample the discrete signal $\mathbf{x}[n]$ by employing fewer cosets while preserving modulation

recognition accuracy. Given that the unknown spectral occupancy typically necessitates $F_t > 2F_l$, we define $R = F_t/2F_l$ as the ratio of the total sampling rate to twice the Landau rate, where a smaller R corresponds to fewer utilized cosets and a larger R implies the use of more. Nevertheless, the spectral aliasing inherent in MCS causes $\mathbf{x}[n]$ to encapsulate complex phase variations and nonlinear coupling characteristics arising from channel-specific time delays, making the direct processing of such unrecovered data tantamount to deconstructing a highly compressed and aliased nonlinear manifold within a high-dimensional space. Further reducing the sampling rate inevitably exacerbates these aliasing and coupling effects, posing significant challenges for both spectrum sensing and modulation recognition tasks.

III. PROPOSED METHOD FOR JOINT SPECTRUM SENSING AND MODULATION RECOGNITION

As depicted in Fig. 1(a), the MCS data are input into the nonlinear embedding module, where they are converted into feature representations. These features are subsequently fed into the spectrum sensing module to extract occupancy information. Finally, features incorporating this spectral context enter the modulation recognition module to output the predicted modulation types. The specific design details of these modules are presented below.

A. Nonlinear Embedding Module

As shown in Fig. 1(b), we first standardize the signal $\mathbf{X}'$ to produce $\mathbf{Y}$ using the following equation:

$$\mathbf{Y} = \frac{\mathbf{X}' - \alpha}{\beta + \epsilon}, \quad (4)$$

where α and β are the mean and standard deviation of $\mathbf{X}$, respectively. A small constant $\epsilon = 10^{-8}$ is added to the denominator to prevent division by zero.

Beyond mere computational compression, the proposed BNE module consolidates irregularly sampled windows to capture local temporal correlations and phase consistencies inherent in MCS. The input signal, $\mathbf{Y} \in \mathbb{R}^{N \times 2P}$, is partitioned into sub-blocks. Each of these sub-blocks, which contains $2PS$ sample points, is then mapped to a single token, compressing the original sequence length from N to $N_{seq} = N/S$. This mapping is implemented using four linear layers, with a GELU activation function and dropout applied after the first three layers. The feature size is transformed from an input of $2PS$ to an intermediate output of $2d_{model}$ by the first three layers, with the final layer producing a compressed feature size of d_{model} . The resulting compressed signal is denoted as $\mathbf{Y}' \in \mathbb{R}^{N_{seq} \times d_{model}}$.

Following the BERT architecture [21], a learnable classification token $e_{cls} \in \mathbb{R}^{1 \times d_{model}}$ is prepended to the sequence. Unlike the data-dependent signal embeddings, e_{cls} is initialized and learned independently of the input data. Positional encodings are then added to the combined token and sequence to form the vector $\mathbf{Z} \in \mathbb{R}^{(N_{seq}+1) \times d_{model}}$, which is then fed into subsequent module.

B. Spectrum Sensing Module

To extract spectral occupancy features, this module utilizes an encoder-only Transformer architecture composed of stacked layers, each comprising a multi-head self-attention (MHA) mechanism and a feed-forward network (FFN). The core self-attention function is defined as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_{model}}}\right)\mathbf{V}, \quad (5)$$

where queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$ are linear projections of the input. MHA parallelizes this process across h heads. For each head i , input $\mathbf{Z}$ is projected via learnable weights $\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V \in \mathbb{R}^{d_{model} \times (d_{model}/h)}$ to generate $\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i$. The final output is obtained by concatenating these heads and applying a linear transformation $\mathbf{W}^O \in \mathbb{R}^{d_{model} \times d_{model}}$:

$$\text{head}_i = \text{Attention}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i), \quad (6)$$

$$\text{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)\mathbf{W}^O. \quad (7)$$

Subsequently, the MHA output is processed by the FFN consisting of two linear layers.

When processing the complex nonlinear manifold structures induced by MCS, the conventional ReLU fails to decouple complex channel-wise higher-order interactions. Therefore, we introduce PolyNorm [22] to explicitly resolve aliasing-induced nonlinear couplings, providing inherent gains over standard linear transformations, as illustrated in Fig. 1(c). Its expression is given by:

$$\text{PolyNorm}(\mathbf{z}) = \sum_{i=0}^r a_i \frac{\mathbf{z}^i}{\|\mathbf{z}^i\|_2}, \quad (8)$$

where $\mathbf{z}^i = [z_1^i, z_2^i, \dots, z_d^i]^\top$ represents element-wise exponentiation, and $\|\cdot\|_2$ denotes the L_2 normalization. This design not only provides high-order nonlinear mapping capabilities

but also functions as a quasi-kernel that maps low-dimensional aliased MCS data into a high-dimensional feature space to disentangle inter-coset couplings. Specifically, PolyNorm theoretically achieves the optimal approximation rate in Sobolev spaces, facilitating the precise fitting of complex nonlinear decision boundaries for spectrum sensing and enabling high-precision detection under undersampling conditions.

Then the e_{cls} token is passed to a spectrum sensing branch. This branch is composed of a fully connected layer (FCL) with a Sigmoid activation function, and its output dimension is equal to the number of subbands it divides into.

C. Modulation Recognition Module

As shown in Fig. 1(a), to reduce computational complexity, we directly utilize the backbone features extracted from the spectrum sensing module for modulation recognition. However, the direct reuse of backbone features renders the deep network susceptible to feature loss and insufficient expressivity. To address this, we employ the PolyNorm within the modulation recognition module, where its characteristic high weight rank and layer-wise diversity effectively suppress feature degradation and over-smoothing in deep Transformers.

To further enhance modulation recognition accuracy, we design composite subband identifiers, uniquely constructed by summing a static sinusoidal encoding with a learnable bias. The static component provides fixed frequency coordinates, analogous to positional encoding, to preserve spectral ordering, while the learnable bias captures subband-specific nonlinear characteristics. Specifically, for a sample with occupied subbands indexed by u and v , we integrate this occupancy information by appending the corresponding identifiers to the backbone features. The identifier pool consists of L tokens, corresponding to the total number of subbands defined in the spectrum sensing process.

Then a new token m_{cls} replaces e_{cls} to extract relevant information. The subsequent processing architecture is analogous to that of the spectrum sensing branch, where the encoded m_{cls} is fed into a classifier to output the modulation types.

IV. EXPERIMENTAL AND EVALUATION

A. Dataset and Experimental Setup

Experiments were conducted using the publicly available GBSense 2022 dataset [23], comprising 102,400 training and 10,000 testing samples acquired via eight multi-coset samplers each operating at 50 million samples per second (MSPS). The spectrum is divided into 24 subbands with a bandwidth of 50 MHz per subband. Since the majority of signals occupy two subbands, the Landau rate F_l is 100 MSPS, establishing a twice Landau rate threshold $2F_l$ of 200 MSPS. To evaluate modulation recognition performance in the challenging regime where the total sampling rate F_t falls below $2F_l$, we specifically utilize only three cosets. This configuration results in a total sampling rate of 150 MSPS, strictly satisfying the sub-Nyquist condition where F_t is less than $2F_l$. All implementations were performed using PyTorch on an NVIDIA GeForce RTX 4090 GPU.

B. Ablation Experiments

We conducted ablation experiments with 3 cosets under sub-Nyquist conditions using a baseline Transformer. As shown in Table I, the BNE module significantly enhances efficiency via signal sequence compression, where the block size S is set to 8, reducing the baseline inference time from 96.50 ms to 18.59 ms. Notably, while the Transformer+PolyNorm model maintained a stable parameter count, its inference time increased to 360.15 ms as the custom activation precluded standard acceleration optimizations. The final proposed BNEPN model successfully mitigated this computational latency through the sequence compression mechanism of the BNE module, optimizing the inference time to 49.82 ms and maintaining high operational efficiency while achieving performance gains. Regarding accuracy, Table I also indicates that PolyNorm plays a dominant role in performance enhancement by effectively alleviating the nonlinear coupling of signals sampled with 3 cosets, thereby elevating both spectrum sensing and modulation recognition accuracy to very high levels. The confusion matrices in Fig. 2 further demonstrate that the method achieves precise spectrum sensing across all subbands, with the primary modulation recognition errors concentrated in the misclassification of QAM128 and QAM256. This is mainly attributed to the dense aliasing effects generated by high-order modulation signals when only 3 cosets are utilized for sub-Nyquist sampling.

TABLE I
Computational complexity, efficiency, and accuracy analysis

Model	Number of Parameters	GPU Memory	Inference Time	Accuracy (Spec./Mod.)
Transformer	25,256,997	1829 MB	96.50 ms	99.45% / 88.88%
Trans.+BNE	26,883,109	382 MB	18.59 ms	99.46% / 88.90%
Trans.+PolyNorm	25,257,029	3111 MB	360.15 ms	99.98% / 91.01%
Proposed BNEPN	26,883,141	550 MB	49.82 ms	99.99% / 91.30%

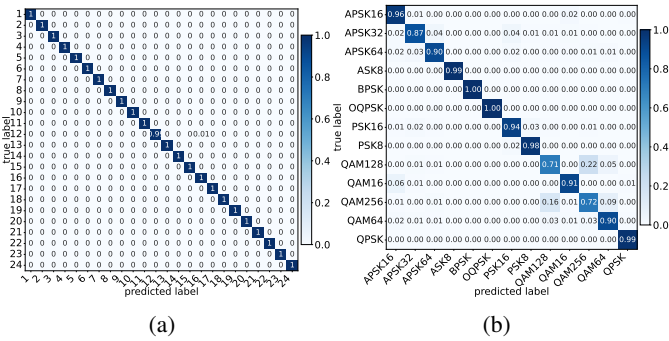


Fig. 2: Confusion matrices of spectrum sensing and modulation recognition using 3 cosets. (a) Confusion matrix of spectrum sensing. (b) Confusion matrix of modulation recognition.

C. Benchmark Comparison

Fig. 3 compares the average modulation recognition accuracy of the proposed BNEPN model with Transformer, ResNet34, and SigChord. Fig. 3(a) details the performance across different sampling rate ratios R , where values of 1.25,

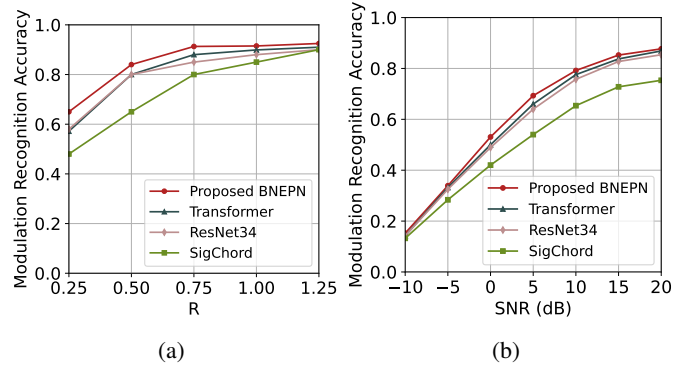


Fig. 3: Modulation recognition performance comparison under different sampling rates and noise levels. (a) Impact of sampling rate ratio R on accuracy. (b) Impact of SNR on accuracy with 3 cosets.

1.00, 0.75, 0.50, and 0.25 correspond to the utilization of 5, 4, 3, 2, and 1 cosets, respectively. Observations indicate that at $R = 0.75$, where the total sampling rate F_t falls below $2F_l$, the BNEPN model maintains performance comparable to the Nyquist-compliant rate of $R = 1.00$. When the total sampling rate is further reduced to $R = 0.50$, BNEPN exhibits a more graceful decline in accuracy compared to other baselines, demonstrating the robustness of PolyNorm against aliasing effects. Fig. 3(b) shows the modulation recognition accuracy with respect to the signal-to-noise ratio (SNR) at $R = 0.75$. It is noted that the GBSense 2022 dataset already contains environmental noise. However, the experiments treat these samples as clean signals before adding additive white Gaussian noise. This methodology results in an effective SNR strictly lower than the nominal value, thereby creating a more rigorous challenge for the models. As depicted in Fig. 3(b), the recognition accuracy for all models exhibits a general upward trend as the SNR increases from -10 dB to 20 dB. The proposed BNEPN consistently outperforms the baseline models across the entire SNR range. In comparison, the standard Transformer performs slightly worse than BNEPN. ResNet34 shows significantly lower accuracy, highlighting the necessity of utilizing spectral information for this task. SigChord demonstrates the sharpest performance degradation under low SNR and limited coset conditions, primarily due to the difficulty of signal recovery in complex noise environments.

V. CONCLUSION

This paper proposes the BNEPN framework for joint wide-band spectrum sensing and modulation recognition in sub-Nyquist systems. The method utilizes the BNE module for efficient signal compression and PolyNorm to resolve aliasing and prevent feature over-smoothing. Additionally, composite subband identifiers are integrated to leverage spectral occupancy information. Experimental results on the GBSense 2022 dataset confirm that BNEPN achieves 99.99% sensing accuracy and superior modulation recognition compared to baselines, even when the sampling rate is below twice the Landau rate. Future work will verify its robustness under real-world mismatches like multi-path fading.

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